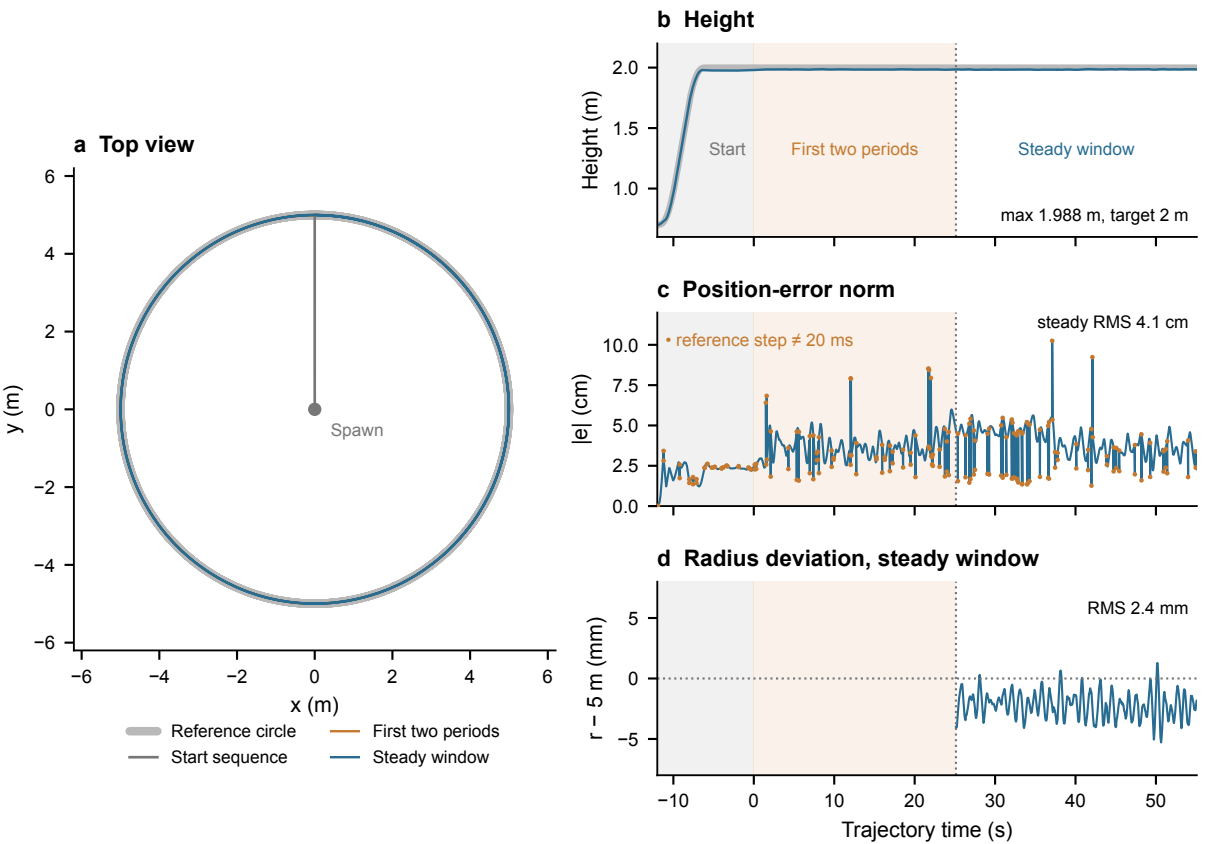

a Top view
Spawn
x (m)
y (m)
Reference circle
Start sequence
First two periods
Steady window
b Height
Height (m)
Start
First two periods
Steady window
max 1.988 m, target 2 m
c Position-error norm
|e| (cm)
reference step ≠ 20 ms
steady RMS 4.1 cm
d Radius deviation, steady window
r − 5 m (mm)
RMS 2.4 mm
Trajectory time (s)